

DATA ANALYTICS MINI PROJECT

Name: Suraj R Bhandare

Roll Number: SA260

Project Title: Global Fuel Price Dynamics: Analyzing the Impact of Crude Oil, Demand, and Geopolitical Events (2015–2026)

Problem Statement:

This project analyzes global petrol and gas prices to understand how factors like crude oil prices, demand, and global events influence fuel prices. The goal is to identify patterns and relationships that affect fuel price fluctuations.

Dataset Description:

Dataset Name: Global Petrol & Gas Price Analysis

Source: Kaggle

Rows: 5860

Columns: 22

Importing Libraries

```
[ ] import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Data Loading

The dataset has been successfully loaded using pandas, and the first 5 rows are displayed to understand its structure.

```
[ ] df = pd.read_csv("petrol_gas_price_analysis_clean.csv")
df.head()
```

	Date	Country	Region	Petrol_Price_USD_per_Liter	Diesel_Price_USD_per_Liter	Natural_Gas_Price_USD_per_MMBtu	Crude_Oil_Price_USD_per_Barrel	Inflation_Rate (%)	Exchange_Rate
0	2015-01-05	Brazil	Latin America	0.675	0.609	1.769	50.91	2.42	
1	2015-01-12	Brazil	Latin America	0.687	0.602	1.565	51.57	3.39	
2	2015-01-19	Brazil	Latin America	0.710	0.642	1.760	53.96	2.03	
3	2015-01-26	Brazil	Latin America	0.802	0.713	1.500	58.69	2.30	
4	2015-02-02	Brazil	Latin America	0.685	0.650	1.500	52.61	1.96	

5 rows × 22 columns

Dataset Shape

The shape of the dataset shows the number of rows and columns. This dataset contains 5860 rows and 22 columns, indicating a sufficiently large dataset for analysis.

```
[ ] df.shape
```

(5860, 22)

Dataset Information

The dataset contains both numerical and categorical columns. It includes variables such as fuel prices, economic indicators, and geopolitical factors.

```
[ ] df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5860 entries, 0 to 5859
Data columns (total 22 columns):
 #   Column                                Non-Null Count  Dtype  
---  --
 0   Date                                5860 non-null   object  
 1   Country                             5860 non-null   object  
 2   Region                              5860 non-null   object  
 3   Petrol_Price_USD_per_Liter          5860 non-null   float64 
 4   Diesel_Price_USD_per_Liter           5860 non-null   float64 
 5   Natural_Gas_Price_USD_per_MMBtu     5860 non-null   float64 
 6   Crude_Oil_Price_USD_per_Barrel       5860 non-null   float64 
 7   Inflation_Rate (%)                   5860 non-null   float64 
 8   Exchange_Rate_vs_USD                 5860 non-null   float64 
 9   GDP_Growth (%)                       5860 non-null   float64 
10   Supply_Index                         5860 non-null   float64 
11   Demand_Index                         5860 non-null   float64 
12   Geopolitical_Risk_Index              5860 non-null   float64 
13   Event_Flag                           5860 non-null   int64  
14   Event_Description                    977 non-null    object  
15   Currency_Devaluation (%)             5860 non-null   float64 
16   Subsidy_Level                       5860 non-null   object  
17   Tax_Rate_on_Fuel (%)                 5860 non-null   float64 
18   Petrol_7d_MA                        5860 non-null   float64 
19   Petrol_28d_vol                       5840 non-null   float64 
20   Petrol_lag_1                        5850 non-null   float64 
21   Crude_lag_1                         5850 non-null   float64 
dtypes: float64(16), int64(1), object(5)
memory usage: 1007.3+ KB
```

Column Datatype

The dataset consists of different data types including integers, floats, and object types. These represent various features like prices, indices, and categorical data such as country and region.

```
[ ] df.dtypes

Date                                object
Country                             object
Region                              object
Petrol_Price_USD_per_Liter          float64
Diesel_Price_USD_per_Liter           float64
Natural_Gas_Price_USD_per_MMBtu     float64
Crude_Oil_Price_USD_per_Barrel       float64
Inflation_Rate (%)                   float64
Exchange_Rate_vs_USD                 float64
GDP_Growth (%)                       float64
Supply_Index                         float64
Demand_Index                         float64
Geopolitical_Risk_Index              float64
Event_Flag                           int64
Event_Description                    object
Currency_Devaluation (%)             float64
Subsidy_Level                       object
Tax_Rate_on_Fuel (%)                 float64
Petrol_7d_MA                        float64
Petrol_28d_vol                       float64
Petrol_lag_1                        float64
Crude_lag_1                         float64

dtype: object
```

Missing Values

Missing values were identified in some columns. These were handled appropriately using filling techniques to ensure data consistency and avoid errors during analysis.

```
[ ] df.isnull().sum()
```

	0
Date	0
Country	0
Region	0
Petrol_Price_USD_per_Liter	0
Diesel_Price_USD_per_Liter	0
Natural_Gas_Price_USD_per_MMBtu	0
Crude_Oil_Price_USD_per_Barrel	0
Inflation_Rate (%)	0
Exchange_Rate_vs_USD	0
GDP_Growth (%)	0
Supply_Index	0
Demand_Index	0
Geopolitical_Risk_Index	0
Event_Flag	0
Event_Description	4883
Currency_Devaluation (%)	0
Subsidy_Level	0
Tax_Rate_on_Fuel (%)	0
Petrol_7d_MA	0
Petrol_28d_vol	20
Petrol_lag_1	10
Crude_lag_1	10

dtype: int64

Data Cleaning

The dataset was checked for missing values and inconsistencies. Some columns contained missing values, especially in event-related and lag features. These issues were handled using appropriate techniques to ensure data quality.

```
[ ]  
  
# Fill Event Description separately  
df['Event_Description'].fillna("No Event", inplace=True)  
df['Petrol_28d_vol'].fillna(df['Petrol_28d_vol'].mean(), inplace=True)  
df['Petrol_lag_1'].fillna(method='bfill', inplace=True)  
df['Crude_lag_1'].fillna(method='bfill', inplace=True)  
  
# Remove duplicates  
df.drop_duplicates(inplace=True)
```

After Data Cleaning

After applying the cleaning techniques, all missing values were successfully handled. The dataset is now free from null values and duplicate records, making it suitable for further analysis and modeling.

```
df.isnull().sum()
```

Date	0
Country	0
Region	0
Petrol_Price_USD_per_Liter	0
Diesel_Price_USD_per_Liter	0
Natural_Gas_Price_USD_per_MMBtu	0
Crude_Oil_Price_USD_per_Barrel	0
Inflation_Rate (%)	0
Exchange_Rate_vs_USD	0
GDP_Growth (%)	0
Supply_Index	0
Demand_Index	0
Geopolitical_Risk_Index	0
Event_Flag	0
Event_Description	0
Currency_Devaluation (%)	0
Subsidy_Level	0
Tax_Rate_on_Fuel (%)	0
Petrol_7d_MA	0
Petrol_28d_vol	0
Petrol_lag_1	0
Crude_lag_1	0

dtype: int64

After applying the cleaning techniques, all missing values were successfully handled. The dataset is now free from null values and duplicate records, making it suitable for further analysis and modeling.

Statistical Analysis

Statistical measures such as mean, median, mode, standard deviation, variance, range, and mid-range were calculated for petrol and crude oil prices.

These measures help in understanding the central tendency and variability of fuel prices. The results show how prices fluctuate over time and indicate the spread and consistency of the data.

```
# Petrol Price Statistics
petrol = df['Petrol_Price_USD_per_Liter']

print("PETROL PRICE STATS")
print("Mean:", petrol.mean())
print("Median:", petrol.median())
print("Mode:", petrol.mode()[0])
print("Standard Deviation:", petrol.std())
print("Variance:", petrol.var())
print("Range:", petrol.max() - petrol.min())
print("Mid Range:", (petrol.max() + petrol.min()) / 2)

# Crude Oil Price Statistics
crude = df['Crude_Oil_Price_USD_per_Barrel']

print("\nCRUDE OIL PRICE STATS")
print("Mean:", crude.mean())
print("Median:", crude.median())
print("Mode:", crude.mode()[0])
print("Standard Deviation:", crude.std())
print("Variance:", crude.var())
print("Range:", crude.max() - crude.min())
print("Mid Range:", (crude.max() + crude.min()) / 2)
```

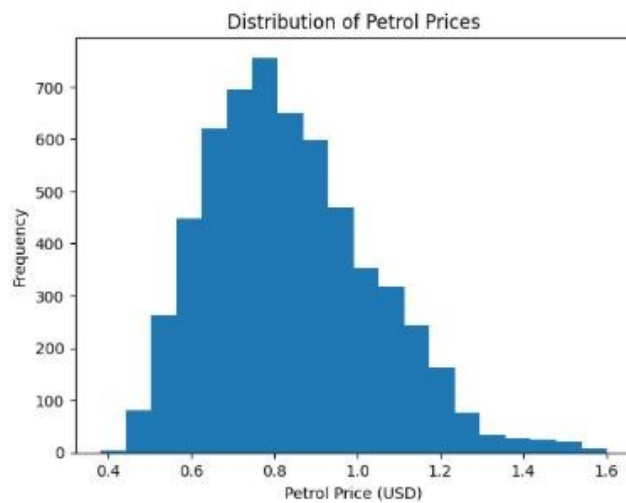
```
PETROL PRICE STATS
Mean: 0.8386
Median: 0.814
Mode: 0.792
Standard Deviation: 0.19953094270934804
Variance: 0.03981259709848113
Range: 1.22
Mid Range: 0.991

CRUDE OIL PRICE STATS
Mean: 57.44340614334471
Median: 56.745000000000005
Mode: 48.47
Standard Deviation: 9.904935734538546
Variance: 98.10775190533863
Range: 56.259999999999999
Mid Range: 63.72
```

Data Visualization

1. Distribution of Global Petrol Prices

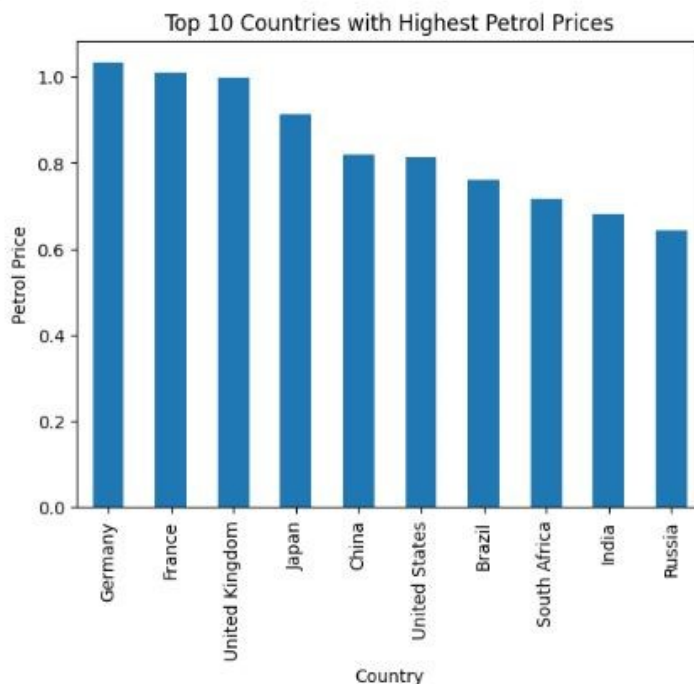
```
plt.hist(df['Petrol_Price_USD_per_Liter'], bins=20)
plt.title("Distribution of Petrol Prices")
plt.xlabel("Petrol Price (USD)")
plt.ylabel("Frequency")
plt.show()
```



The histogram shows that most petrol prices fall within a moderate range, with fewer countries having very high prices. This indicates that fuel prices are generally stable, with some variations across regions.

2. Top 10 Countries with Highest Petrol Prices

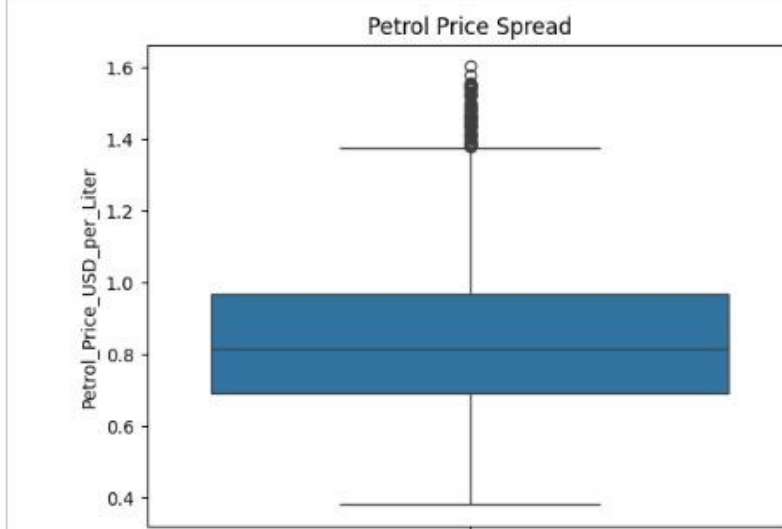
```
top10 = df.groupby('Country')['Petrol_Price_USD_per_Liter'].mean().sort_values(ascending=False).head(10)
top10.plot(kind='bar')
plt.title("Top 10 Countries with Highest Petrol Prices")
plt.xlabel("Country")
plt.ylabel("Petrol Price")
plt.show()
```



This bar chart shows the top 10 countries with the highest petrol prices. It highlights that fuel prices vary across countries, with some countries having significantly higher prices than others.

3. Petrol Price Distribution and Outliers

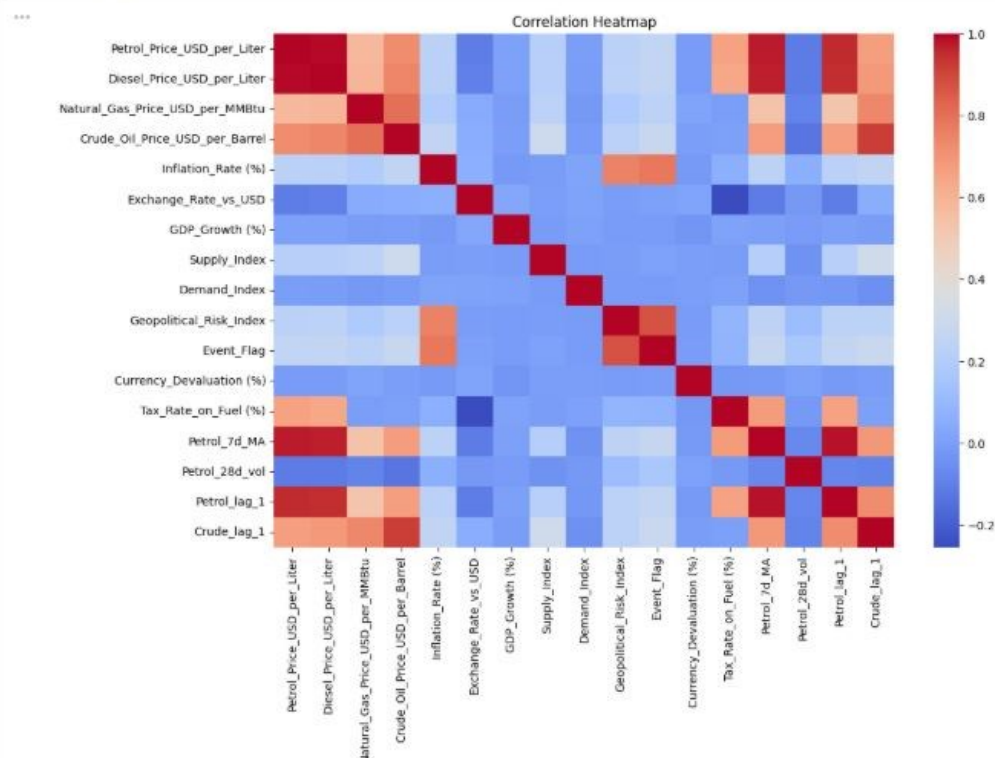
```
[ ] sns.boxplot(y=df['Petrol_Price_USD_per_Liter'])  
plt.title("Petrol Price Spread")  
plt.show()
```



The boxplot shows the distribution of petrol prices and highlights the presence of outliers. Most prices fall within a normal range, while some higher values indicate sudden increases in fuel prices.

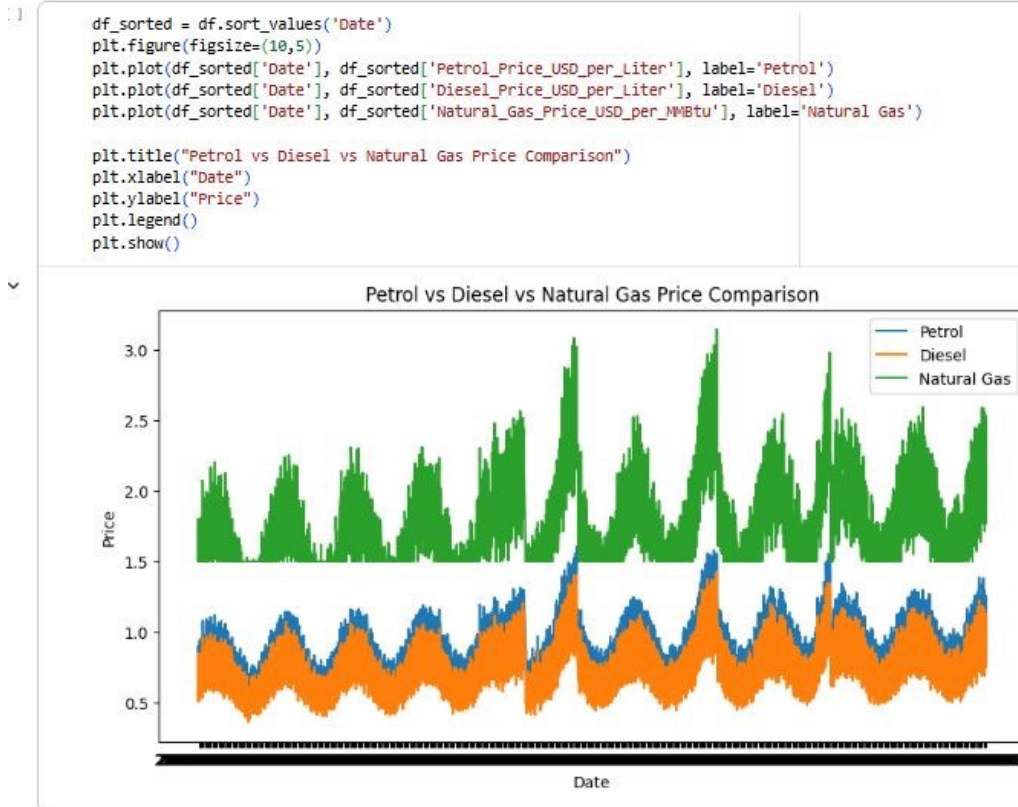
4. Petrol Price Distribution and Outliers

```
[ ] plt.figure(figsize=(12,8))  
sns.heatmap(df.corr(numeric_only=True), cmap='coolwarm')  
plt.title("Correlation Heatmap")  
plt.show()
```



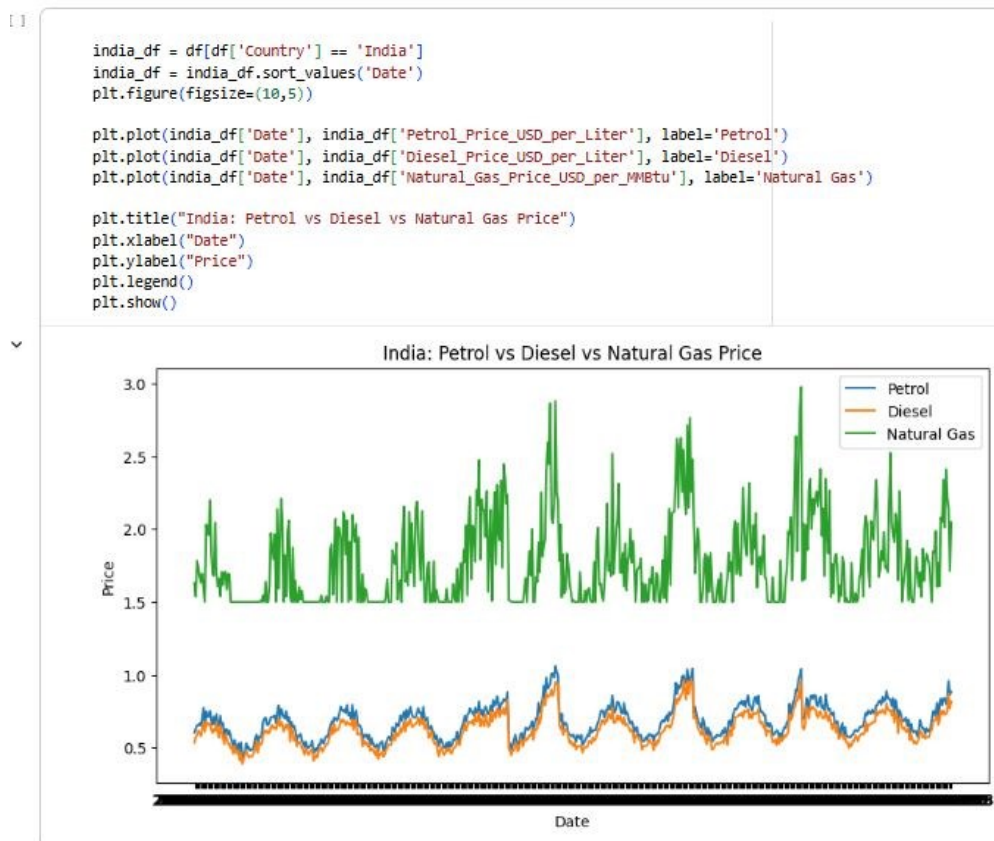
The heatmap shows the relationship between different variables. Petrol prices have a strong positive correlation with crude oil prices and diesel prices, indicating that these factors significantly influence fuel prices.

5. Comparative Analysis of Petrol, Diesel, and Natural Gas Prices



This graph compares petrol, diesel, and natural gas prices over time. Petrol and diesel follow a similar trend, while natural gas shows higher fluctuations, indicating different market behavior.

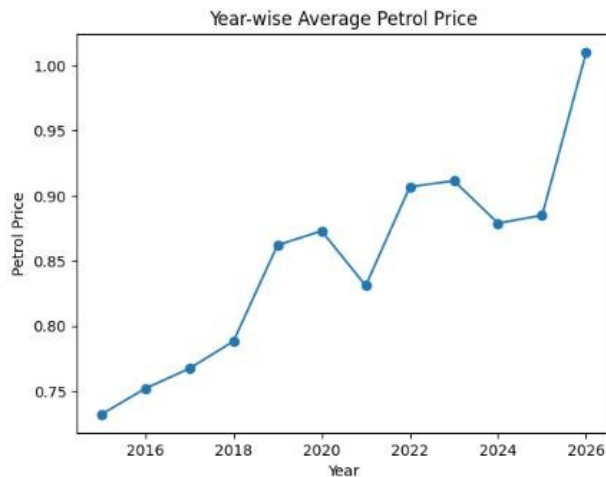
6. Fuel Price Trends in India



This graph shows fuel price trends in India. Petrol prices are generally higher than diesel, and both follow a similar pattern, while natural gas shows more fluctuations.

7. Year-wise Trend of Petrol Prices

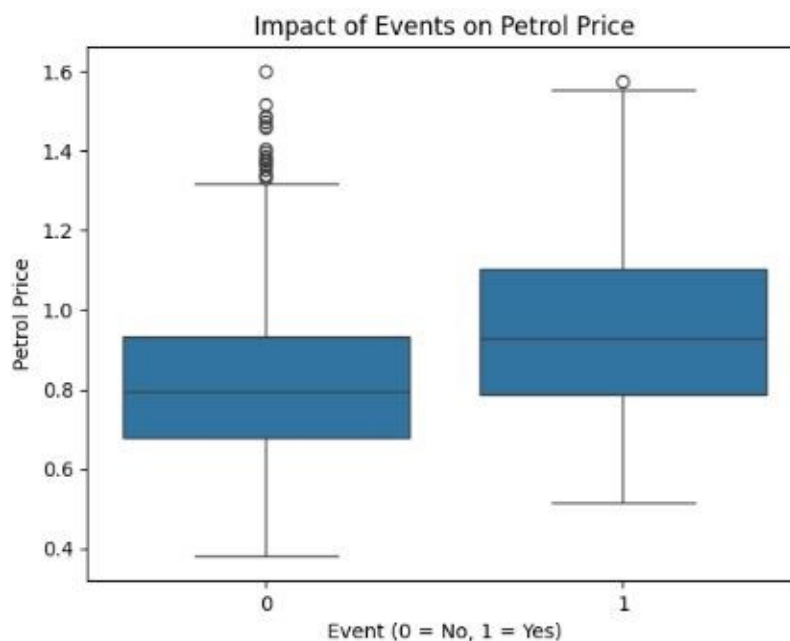
```
[ ] df['Year'] = pd.to_datetime(df['Date']).dt.year  
  
year_avg = df.groupby('Year')['Petrol_Price_USD_per_Liter'].mean()  
  
year_avg.plot(kind='line', marker='o')  
plt.title("Year-wise Average Petrol Price")  
plt.xlabel("Year")  
plt.ylabel("Petrol Price")  
plt.show()
```



This graph shows the yearly trend of petrol prices. It indicates that petrol prices have generally increased over time, reflecting rising demand and economic factors.

8. Impact of Global Events on Petrol Prices

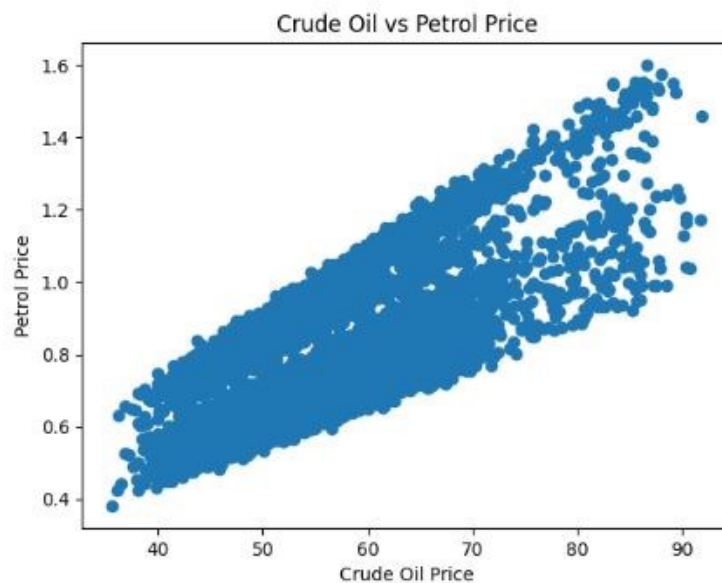
```
[ ] sns.boxplot(x='Event_Flag', y='Petrol_Price_USD_per_Liter', data=df)  
plt.title("Impact of Events on Petrol Price")  
plt.xlabel("Event (0 = No, 1 = Yes)")  
plt.ylabel("Petrol Price")  
plt.show()
```



This graph compares petrol prices during normal periods and during global events. It shows that prices tend to be higher during events, indicating the impact of external factors on fuel prices.

9. Relationship Between Crude Oil and Petrol Prices

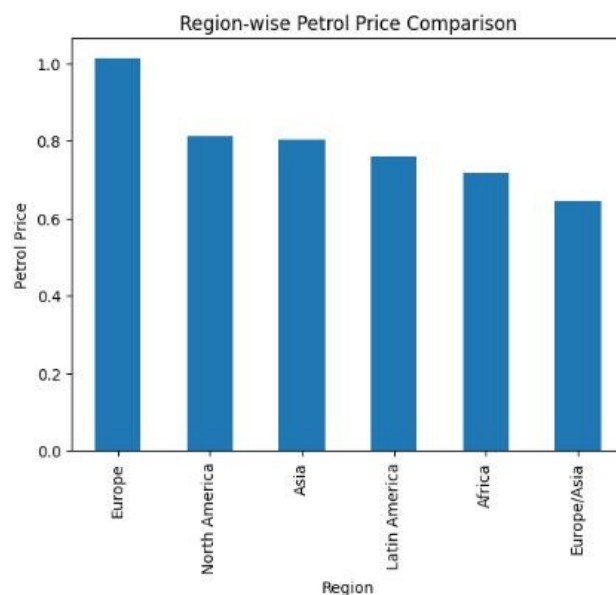
```
[ ] plt.scatter(df['Crude_Oil_Price_USD_per_Barrel'], df['Petrol_Price_USD_per_Liter'])
plt.title("Crude Oil vs Petrol Price")
plt.xlabel("Crude Oil Price")
plt.ylabel("Petrol Price")
plt.show()
```



This scatter plot shows a positive relationship between crude oil prices and petrol prices. As crude oil prices increase, petrol prices also tend to increase.

10. Regional Comparison of Petrol Prices

```
[ ] region_avg = df.groupby('Region')['Petrol_Price_USD_per_Liter'].mean().sort_values(ascending=False)
region_avg.plot(kind='bar')
plt.title("Region-wise Petrol Price Comparison")
plt.xlabel("Region")
plt.ylabel("Petrol Price")
plt.show()
```



This graph compares average petrol prices across different regions. It shows that fuel prices vary by region, with some regions having higher prices due to economic conditions and fuel policies.

11. Simple Model

```
[ ] from sklearn.model_selection import train_test_split
    from sklearn.linear_model import LinearRegression

    df_ml = df.copy()

    # Remove problematic columns
    df_ml = df_ml.drop(columns=['Date', 'Event_Description', 'Month'], errors='ignore')

    # Convert categorical to numeric
    df_ml = pd.get_dummies(df_ml, drop_first=True)

    # Remove missing values
    df_ml = df_ml.dropna()

    # Features & Target
    X = df_ml.drop('Petrol_Price_USD_per_Liter', axis=1)
    y = df_ml['Petrol_Price_USD_per_Liter']

    # Split
    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

    # Model
    model = LinearRegression()
    model.fit(X_train, y_train)

    # Accuracy
    print("Model Accuracy:", model.score(X_test, y_test))

Model Accuracy: 0.9896663387289769
```

The model achieved a high accuracy score, indicating that it can effectively predict petrol prices based on the given features. However, some features are closely related to petrol prices, which may lead to very high accuracy. Overall, the model demonstrates a strong relationship between input variables and petrol prices.

Insights

1. The histogram shows that most petrol prices are concentrated in a moderate range, indicating that fuel prices are generally stable with a few high-price outliers.
2. The bar chart of top 10 countries shows that certain countries have significantly higher petrol prices, suggesting the impact of higher taxes and import dependency.
3. The boxplot shows the spread of petrol prices and highlights the presence of outliers, indicating sudden spikes in prices in some cases.
4. The correlation heatmap shows a strong positive relationship between petrol prices and crude oil prices, indicating that crude oil is a major influencing factor.
5. The comparison graph of petrol, diesel, and natural gas shows that petrol and diesel follow similar trends, while natural gas is more volatile.
6. The India-specific graph shows that petrol prices are consistently higher than diesel, and both follow a similar trend over time.
7. The year-wise trend graph shows a gradual increase in petrol prices over the years, indicating rising demand and economic impact.
8. The event impact boxplot shows that petrol prices are generally higher during global events, indicating the influence of geopolitical factors.
9. The scatter plot between crude oil and petrol prices shows a strong positive correlation, confirming that petrol prices increase with crude oil prices.

10. The region-wise bar chart shows that petrol prices vary across regions, indicating differences in economic conditions, taxation, and fuel policies.

Recommendations

1. Governments should regulate fuel taxes during periods of rising crude oil prices to reduce the financial burden on consumers.
2. Countries should invest in alternative and renewable energy sources to reduce dependence on crude oil and stabilize fuel prices.
3. Countries should maintain strategic fuel reserves to handle sudden price spikes caused by global events or supply disruptions.

Github link : <https://github.com/Surajbhandare07/global-fuel-price-analysis>